

Emanuel Roth

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Date & Place of Birth | *October 9 1998, Brunei*

Address | *Edinburgh, United Kingdom*

Languages | *Human: Fluent in English and German, basic command in Tagalog and Chinese. Computer: Python, Latex, SAGEmath*

Citizenship | *German, Filipino*

Education

University of Edinburgh | *PhD Student in Mathematics* **2024 - 2028**

- Mathematics PhD supervised by Johan Martens **2024 - 2028**
Topic: A symplectic approach to global Springer theory (the symplectic geometry of moduli spaces of parahoric Higgs bundles)

University of Heidelberg | *Bachelors and Masters Student in Mathematics* **2017 - 2024**

- Mathematics Master supervised by Florent Schaffhauser: *Note 1,7* **2022 - 2024**
Topic: “Harder-Narasimhan filtrations of decorated vector bundles”.
- Mathematics Master Erasmus exchange student: *University of Vienna* **Winter Semester 2022 - 2023**
- Mathematics Bachelor: *Note 2,1* **2017 - 2022**
Topic: “The Mitchell embedding theorem”

Teaching

Further Complex Variables | *University of Edinburgh* **Winter Semester 2025**

General Topology | *University of Edinburgh* **Winter Semester 2025**

Honours Analysis - Skills | *University of Edinburgh* **Winter Semester 2025**

Proofs and Problem Solving | *University of Edinburgh* **Summer Semester 2025**

Fundamentals of Pure Mathematics | *University of Edinburgh* **Winter Semester 2024**

Mathematics for Physicists 3 | *University of Heidelberg* **Winter Semester 2023 - 2024**

Mathematics for Physicists 2 | *University of Heidelberg* **Summer Semester 2023**

Linear Algebra 2 | *University of Heidelberg* **Summer Semester 2022**

Linear Algebra 1 | *University of Heidelberg* **Winter Semester 2021 - 2022**

Preliminary Courses | *University of Heidelberg* **October 2021**

Preliminary Courses | *University of Heidelberg* **October 2020**

Preprints

Harder-Narasimhan Filtrations of Decorated Vector Bundles - with Florent Schaffhauser - [arxiv:2511.17371](https://arxiv.org/abs/2511.17371)

Selection of Conferences and Visits

Enumerative Geometry and Geometric Representation Theory | *Università di Pisa* **June 2025**

Spring School: Group Actions and Symplectic Singularities | *University of Lille* **June 2025**

Stratifications of Higgs bundle moduli spaces | *Universidade de Santiago de Compostela* **May 2025**

Research Visit, meeting Florent Schaffhauser | *University of Heidelberg* **February 2025**

Winter School on Algebraic Geometry | *Lancaster University* **16 - 20 December 2024**

Moduli Spaces of Vector Bundles and Curves | *Université de Cergy-Pontoise* **11 - 13 December 2023**

Talks and Seminars

- Organized AGQ reading group on stacks.
- Given talks in AGQ reading groups: Geometric invariant theory, Enumerative geometry, Geometric representation theory, stacks.
- Talk in AGQ Conference and GEARS (Glasgow Edinburgh Algebra Research Student Seminar) December 2025.